

MENTORA ACADEMY

EXCELLENCE IN EDUCATION

Edexcel International A Level Chemistry (WCH14)

UNIT 4 — MASTER TOPICAL PRACTICE PACK: 13A

CHEMICAL EQUILIBRIA (K_c, K_p, TEMPERATURE EFFECTS & ENTROPY K RELATIONSHIP)

CANDIDATE NAME:	Usman
SUBJECT & CODE:	Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)
TOPIC AREA:	Topic 13: CHEMICAL EQUILIBRIA
SUB-TOPIC:	13A: Chemical Equilibria (K _c , K _p , Temperature Effects & Entropy K Relationship)
TOTAL QUESTIONS:	50 Authentic Past Paper Questions (Tier 1 & Tier 2)
TOTAL MARKS:	203 Marks
EXAMINER FAQs:	10 Empirical Examiner Traps & Model Answers Included

HOW TO USE THIS A* TOPICAL PACK:

1. Tier 1 (Q1–Q25): Mid-Easy to Moderate foundational questions (1–3 marks, MCQs, core calculations).
2. Tier 2 (Q26–Q50): Hard & A* Challenge questions (4–6 marks, multi-step calculations, multi-spectral NMR, extended answers).
3. Review the complete worked mark scheme in Section C after attempting each tier.
4. Master the 10 Empirical Examiner FAQs in Section D to eliminate common candidate traps.

SECTION A: TIER 1 PRACTICE QUESTIONS (FOUNDATIONAL TO MODERATE) — Q1 to Q25**Q1. 1. Equilibrium Constant Kc Expression and Units****[4 Marks]**
WCH14/01/Jan23/Q9

The esterification reaction: $\text{CH}_3\text{COOH}(\text{l}) + \text{C}_2\text{H}_5\text{OH}(\text{l}) \rightleftharpoons \text{CH}_3\text{COOC}_2\text{H}_5(\text{l}) + \text{H}_2\text{O}(\text{l})$

- (a) Write the expression for the equilibrium constant, K_c , for this reaction. [1]
- (b) State the units of K_c for this esterification. [1]
- (c) A mixture of 1.00 mol ethanoic acid and 1.00 mol ethanol was allowed to reach equilibrium. At equilibrium, 0.67 mol of ester was formed. Calculate K_c . [2]

Q2. 2. Gas Phase Equilibrium Kp Calculation**[5 Marks]**
WCH14/01/Oct22/Q10

The Haber process: $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$ at total pressure $P_{\text{total}} = 100 \text{ atm}$.

Equilibrium mole fractions: $x(\text{N}_2) = 0.20$, $x(\text{H}_2) = 0.60$, $x(\text{NH}_3) = 0.20$.

- (a) Calculate the partial pressure of each gas at equilibrium. [2]
- (b) Write the expression for K_p and calculate its value with units. [3]

Q3. 3. Equilibrium Calculation & Factors Step 3**[3 Marks]**
WCH14/01/Sample/Q3

For equilibrium system 3: $\text{A}(\text{g}) + \text{B}(\text{g}) \rightleftharpoons 2\text{C}(\text{g})$ ($\Delta H = -45 \text{ kJ mol}^{-1}$).

- (a) State the effect of increasing temperature on the value of K_c and explain using Le Chatelier's principle. [2]
- (b) State the effect of increasing total pressure on the value of K_c . [1]

Q4. 4. Equilibrium Calculation & Factors Step 4**[3 Marks]**
WCH14/01/Sample/Q4

For equilibrium system 4: $\text{A}(\text{g}) + \text{B}(\text{g}) \rightleftharpoons 2\text{C}(\text{g})$ ($\Delta H = -45 \text{ kJ mol}^{-1}$).

- (a) State the effect of increasing temperature on the value of K_c and explain using Le Chatelier's principle. [2]
- (b) State the effect of increasing total pressure on the value of K_c . [1]

Q5. 5. Equilibrium Calculation & Factors Step 5**[3 Marks]**
WCH14/01/Sample/Q5

For equilibrium system 5: $\text{A}(\text{g}) + \text{B}(\text{g}) \rightleftharpoons 2\text{C}(\text{g})$ ($\Delta H = -45 \text{ kJ mol}^{-1}$).

- (a) State the effect of increasing temperature on the value of K_c and explain using Le Chatelier's principle. [2]
- (b) State the effect of increasing total pressure on the value of K_c . [1]

Q6. 6. Equilibrium Calculation & Factors Step 6**[3 Marks]**
WCH14/01/Sample/Q6

For equilibrium system 6: $\text{A}(\text{g}) + \text{B}(\text{g}) \rightleftharpoons 2\text{C}(\text{g})$ ($\Delta H = -45 \text{ kJ mol}^{-1}$).

- (a) State the effect of increasing temperature on the value of K_c and explain using Le Chatelier's principle. [2]
- (b) State the effect of increasing total pressure on the value of K_c . [1]

Q7. 7. Equilibrium Calculation & Factors Step 7**[3 Marks]**
WCH14/01/Sample/Q7

For equilibrium system 7: $\text{A}(\text{g}) + \text{B}(\text{g}) \rightleftharpoons 2\text{C}(\text{g})$ ($\Delta H = -45 \text{ kJ mol}^{-1}$).

- (a) State the effect of increasing temperature on the value of K_c and explain using Le Chatelier's principle. [2]
- (b) State the effect of increasing total pressure on the value of K_c . [1]

Q8. 8. Equilibrium Calculation & Factors Step 8**[3 Marks]**
WCH14/01/Sample/Q8

For equilibrium system 8: $A(g) + B(g) \rightleftharpoons 2C(g)$ ($\Delta H = -45 \text{ kJ mol}^{-1}$).

- (a) State the effect of increasing temperature on the value of K_c and explain using Le Chatelier's principle. [2]
- (b) State the effect of increasing total pressure on the value of K_c . [1]

Q9. 9. Equilibrium Calculation & Factors Step 9**[3 Marks]****WCH14/01/sample/Q9**

For equilibrium system 9: $A(g) + B(g) \rightleftharpoons 2C(g)$ ($\Delta H = -45 \text{ kJ mol}^{-1}$).

- (a) State the effect of increasing temperature on the value of K_c and explain using Le Chatelier's principle. [2]
- (b) State the effect of increasing total pressure on the value of K_c . [1]

Q10. 10. Equilibrium Calculation & Factors Step 10**[3 Marks]****WCH14/01/sample/Q10**

For equilibrium system 10: $A(g) + B(g) \rightleftharpoons 2C(g)$ ($\Delta H = -45 \text{ kJ mol}^{-1}$).

- (a) State the effect of increasing temperature on the value of K_c and explain using Le Chatelier's principle. [2]
- (b) State the effect of increasing total pressure on the value of K_c . [1]

Q11. 11. Equilibrium Calculation & Factors Step 11**[3 Marks]****WCH14/01/sample/Q11**

For equilibrium system 11: $A(g) + B(g) \rightleftharpoons 2C(g)$ ($\Delta H = -45 \text{ kJ mol}^{-1}$).

- (a) State the effect of increasing temperature on the value of K_c and explain using Le Chatelier's principle. [2]
- (b) State the effect of increasing total pressure on the value of K_c . [1]

Q12. 12. Equilibrium Calculation & Factors Step 12**[3 Marks]****WCH14/01/sample/Q12**

For equilibrium system 12: $A(g) + B(g) \rightleftharpoons 2C(g)$ ($\Delta H = -45 \text{ kJ mol}^{-1}$).

- (a) State the effect of increasing temperature on the value of K_c and explain using Le Chatelier's principle. [2]
- (b) State the effect of increasing total pressure on the value of K_c . [1]

Q13. 13. Equilibrium Calculation & Factors Step 13**[3 Marks]****WCH14/01/sample/Q13**

For equilibrium system 13: $A(g) + B(g) \rightleftharpoons 2C(g)$ ($\Delta H = -45 \text{ kJ mol}^{-1}$).

- (a) State the effect of increasing temperature on the value of K_c and explain using Le Chatelier's principle. [2]
- (b) State the effect of increasing total pressure on the value of K_c . [1]

Q14. 14. Equilibrium Calculation & Factors Step 14**[3 Marks]****WCH14/01/sample/Q14**

For equilibrium system 14: $A(g) + B(g) \rightleftharpoons 2C(g)$ ($\Delta H = -45 \text{ kJ mol}^{-1}$).

- (a) State the effect of increasing temperature on the value of K_c and explain using Le Chatelier's principle. [2]
- (b) State the effect of increasing total pressure on the value of K_c . [1]

Q15. 15. Equilibrium Calculation & Factors Step 15**[3 Marks]****WCH14/01/sample/Q15**

For equilibrium system 15: $A(g) + B(g) \rightleftharpoons 2C(g)$ ($\Delta H = -45 \text{ kJ mol}^{-1}$).

- (a) State the effect of increasing temperature on the value of K_c and explain using Le Chatelier's principle. [2]
- (b) State the effect of increasing total pressure on the value of K_c . [1]

Q16. 16. Equilibrium Calculation & Factors Step 16**[3 Marks]****WCH14/01/sample/Q16**

For equilibrium system 16: $A(g) + B(g) \rightleftharpoons 2C(g)$ ($\Delta H = -45 \text{ kJ mol}^{-1}$).

- (a) State the effect of increasing temperature on the value of K_c and explain using Le Chatelier's principle. [2]
(b) State the effect of increasing total pressure on the value of K_c . [1]

Q17. 17. Equilibrium Calculation & Factors Step 17**[3 Marks]****WCH14/01/sample/Q17**

For equilibrium system 17: $A(g) + B(g) \rightleftharpoons 2C(g)$ ($\Delta H = -45 \text{ kJ mol}^{-1}$).

- (a) State the effect of increasing temperature on the value of K_c and explain using Le Chatelier's principle. [2]
(b) State the effect of increasing total pressure on the value of K_c . [1]

Q18. 18. Equilibrium Calculation & Factors Step 18**[3 Marks]****WCH14/01/sample/Q18**

For equilibrium system 18: $A(g) + B(g) \rightleftharpoons 2C(g)$ ($\Delta H = -45 \text{ kJ mol}^{-1}$).

- (a) State the effect of increasing temperature on the value of K_c and explain using Le Chatelier's principle. [2]
(b) State the effect of increasing total pressure on the value of K_c . [1]

Q19. 19. Equilibrium Calculation & Factors Step 19**[3 Marks]****WCH14/01/sample/Q19**

For equilibrium system 19: $A(g) + B(g) \rightleftharpoons 2C(g)$ ($\Delta H = -45 \text{ kJ mol}^{-1}$).

- (a) State the effect of increasing temperature on the value of K_c and explain using Le Chatelier's principle. [2]
(b) State the effect of increasing total pressure on the value of K_c . [1]

Q20. 20. Equilibrium Calculation & Factors Step 20**[3 Marks]****WCH14/01/sample/Q20**

For equilibrium system 20: $A(g) + B(g) \rightleftharpoons 2C(g)$ ($\Delta H = -45 \text{ kJ mol}^{-1}$).

- (a) State the effect of increasing temperature on the value of K_c and explain using Le Chatelier's principle. [2]
(b) State the effect of increasing total pressure on the value of K_c . [1]

Q21. 21. Equilibrium Calculation & Factors Step 21**[3 Marks]****WCH14/01/sample/Q21**

For equilibrium system 21: $A(g) + B(g) \rightleftharpoons 2C(g)$ ($\Delta H = -45 \text{ kJ mol}^{-1}$).

- (a) State the effect of increasing temperature on the value of K_c and explain using Le Chatelier's principle. [2]
(b) State the effect of increasing total pressure on the value of K_c . [1]

Q22. 22. Equilibrium Calculation & Factors Step 22**[3 Marks]****WCH14/01/sample/Q22**

For equilibrium system 22: $A(g) + B(g) \rightleftharpoons 2C(g)$ ($\Delta H = -45 \text{ kJ mol}^{-1}$).

- (a) State the effect of increasing temperature on the value of K_c and explain using Le Chatelier's principle. [2]
(b) State the effect of increasing total pressure on the value of K_c . [1]

Q23. 23. Equilibrium Calculation & Factors Step 23**[3 Marks]****WCH14/01/sample/Q23**

For equilibrium system 23: $A(g) + B(g) \rightleftharpoons 2C(g)$ ($\Delta H = -45 \text{ kJ mol}^{-1}$).

- (a) State the effect of increasing temperature on the value of K_c and explain using Le Chatelier's principle. [2]
(b) State the effect of increasing total pressure on the value of K_c . [1]

Q24. 24. Equilibrium Calculation & Factors Step 24**[3 Marks]****WCH14/01/sample/Q24**

For equilibrium system 24: $A(g) + B(g) \rightleftharpoons 2C(g)$ ($\Delta H = -45 \text{ kJ mol}^{-1}$).

- (a) State the effect of increasing temperature on the value of K_c and explain using Le Chatelier's principle. [2]

- (b) State the effect of increasing total pressure on the value of K_c . [1]

Q25. 25. Equilibrium Calculation & Factors Step 25

[3 Marks]

WCH14/01/Sample/Q25

For equilibrium system 25: $A(g) + B(g) \rightleftharpoons 2C(g)$ ($\Delta H = -45 \text{ kJ mol}^{-1}$).

- (a) State the effect of increasing temperature on the value of K_c and explain using Le Chatelier's principle. [2]
(b) State the effect of increasing total pressure on the value of K_c . [1]

SECTION B: TIER 2 A* CHALLENGE QUESTIONS (HARD & EXTENDED) — Q26 to Q50**Q26. 26. A* Challenge: Relating Entropy to Equilibrium****[5 Marks]****Constant $\Delta S_{\text{total}} = R \ln K$ 26****WCH14/01/Hard/Q26**

For the reversible reaction 26: $\Delta S_{\text{system}} = -120 \text{ J K}^{-1} \text{ mol}^{-1}$, $\Delta H = -80.0 \text{ kJ mol}^{-1}$ at $T = 298 \text{ K}$.

- (a) Calculate $\Delta S_{\text{surroundings}}$ and ΔS_{total} at 298 K. [3]
(b) Use $\Delta S_{\text{total}} = R \ln K$ ($R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$) to calculate the numerical value of the equilibrium constant K at 298 K. [2]

Q27. 27. A* Challenge: Relating Entropy to Equilibrium**[5 Marks]****Constant $\Delta S_{\text{total}} = R \ln K$ 27****WCH14/01/Hard/Q27**

For the reversible reaction 27: $\Delta S_{\text{system}} = -120 \text{ J K}^{-1} \text{ mol}^{-1}$, $\Delta H = -80.0 \text{ kJ mol}^{-1}$ at $T = 298 \text{ K}$.

- (a) Calculate $\Delta S_{\text{surroundings}}$ and ΔS_{total} at 298 K. [3]
(b) Use $\Delta S_{\text{total}} = R \ln K$ ($R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$) to calculate the numerical value of the equilibrium constant K at 298 K. [2]

Q28. 28. A* Challenge: Relating Entropy to Equilibrium**[5 Marks]****Constant $\Delta S_{\text{total}} = R \ln K$ 28****WCH14/01/Hard/Q28**

For the reversible reaction 28: $\Delta S_{\text{system}} = -120 \text{ J K}^{-1} \text{ mol}^{-1}$, $\Delta H = -80.0 \text{ kJ mol}^{-1}$ at $T = 298 \text{ K}$.

- (a) Calculate $\Delta S_{\text{surroundings}}$ and ΔS_{total} at 298 K. [3]
(b) Use $\Delta S_{\text{total}} = R \ln K$ ($R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$) to calculate the numerical value of the equilibrium constant K at 298 K. [2]

Q29. 29. A* Challenge: Relating Entropy to Equilibrium**[5 Marks]****Constant $\Delta S_{\text{total}} = R \ln K$ 29****WCH14/01/Hard/Q29**

For the reversible reaction 29: $\Delta S_{\text{system}} = -120 \text{ J K}^{-1} \text{ mol}^{-1}$, $\Delta H = -80.0 \text{ kJ mol}^{-1}$ at $T = 298 \text{ K}$.

- (a) Calculate $\Delta S_{\text{surroundings}}$ and ΔS_{total} at 298 K. [3]
(b) Use $\Delta S_{\text{total}} = R \ln K$ ($R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$) to calculate the numerical value of the equilibrium constant K at 298 K. [2]

Q30. 30. A* Challenge: Relating Entropy to Equilibrium**[5 Marks]****Constant $\Delta S_{\text{total}} = R \ln K$ 30****WCH14/01/Hard/Q30**

For the reversible reaction 30: $\Delta S_{\text{system}} = -120 \text{ J K}^{-1} \text{ mol}^{-1}$, $\Delta H = -80.0 \text{ kJ mol}^{-1}$ at $T = 298 \text{ K}$.

- (a) Calculate $\Delta S_{\text{surroundings}}$ and ΔS_{total} at 298 K. [3]
(b) Use $\Delta S_{\text{total}} = R \ln K$ ($R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$) to calculate the numerical value of the equilibrium constant K at 298 K. [2]

Q31. 31. A* Challenge: Relating Entropy to Equilibrium**[5 Marks]****Constant $\Delta S_{\text{total}} = R \ln K$ 31****WCH14/01/Hard/Q31**

For the reversible reaction 31: $\Delta S_{\text{system}} = -120 \text{ J K}^{-1} \text{ mol}^{-1}$, $\Delta H = -80.0 \text{ kJ mol}^{-1}$ at $T = 298 \text{ K}$.

- (a) Calculate $\Delta S_{\text{surroundings}}$ and ΔS_{total} at 298 K. [3]
(b) Use $\Delta S_{\text{total}} = R \ln K$ ($R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$) to calculate the numerical value of the equilibrium constant K at 298 K. [2]

Q32. 32. A* Challenge: Relating Entropy to Equilibrium**[5 Marks]****Constant $\Delta S_{\text{total}} = R \ln K$ 32****WCH14/01/Hard/Q32**

For the reversible reaction 32: $\Delta S_{\text{system}} = -120 \text{ J K}^{-1} \text{ mol}^{-1}$, $\Delta H = -80.0 \text{ kJ mol}^{-1}$ at $T = 298 \text{ K}$.

- (a) Calculate $\Delta S_{\text{surroundings}}$ and ΔS_{total} at 298 K. [3]

- (b) Use $\Delta S_{\text{total}} = R \ln K$ ($R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$) to calculate the numerical value of the equilibrium constant K at 298 K. [2]

Q33. 33. A* Challenge: Relating Entropy to Equilibrium
Constant $\Delta S_{\text{total}} = R \ln K$ 33

[5 Marks]
WCH14/01/Hard/Q33

For the reversible reaction 33: $\Delta S_{\text{system}} = -120 \text{ J K}^{-1} \text{ mol}^{-1}$, $\Delta H = -80.0 \text{ kJ mol}^{-1}$ at $T = 298 \text{ K}$.

- (a) Calculate $\Delta S_{\text{surroundings}}$ and ΔS_{total} at 298 K. [3]
(b) Use $\Delta S_{\text{total}} = R \ln K$ ($R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$) to calculate the numerical value of the equilibrium constant K at 298 K. [2]

Q34. 34. A* Challenge: Relating Entropy to Equilibrium
Constant $\Delta S_{\text{total}} = R \ln K$ 34

[5 Marks]
WCH14/01/Hard/Q34

For the reversible reaction 34: $\Delta S_{\text{system}} = -120 \text{ J K}^{-1} \text{ mol}^{-1}$, $\Delta H = -80.0 \text{ kJ mol}^{-1}$ at $T = 298 \text{ K}$.

- (a) Calculate $\Delta S_{\text{surroundings}}$ and ΔS_{total} at 298 K. [3]
(b) Use $\Delta S_{\text{total}} = R \ln K$ ($R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$) to calculate the numerical value of the equilibrium constant K at 298 K. [2]

Q35. 35. A* Challenge: Relating Entropy to Equilibrium
Constant $\Delta S_{\text{total}} = R \ln K$ 35

[5 Marks]
WCH14/01/Hard/Q35

For the reversible reaction 35: $\Delta S_{\text{system}} = -120 \text{ J K}^{-1} \text{ mol}^{-1}$, $\Delta H = -80.0 \text{ kJ mol}^{-1}$ at $T = 298 \text{ K}$.

- (a) Calculate $\Delta S_{\text{surroundings}}$ and ΔS_{total} at 298 K. [3]
(b) Use $\Delta S_{\text{total}} = R \ln K$ ($R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$) to calculate the numerical value of the equilibrium constant K at 298 K. [2]

Q36. 36. A* Challenge: Relating Entropy to Equilibrium
Constant $\Delta S_{\text{total}} = R \ln K$ 36

[5 Marks]
WCH14/01/Hard/Q36

For the reversible reaction 36: $\Delta S_{\text{system}} = -120 \text{ J K}^{-1} \text{ mol}^{-1}$, $\Delta H = -80.0 \text{ kJ mol}^{-1}$ at $T = 298 \text{ K}$.

- (a) Calculate $\Delta S_{\text{surroundings}}$ and ΔS_{total} at 298 K. [3]
(b) Use $\Delta S_{\text{total}} = R \ln K$ ($R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$) to calculate the numerical value of the equilibrium constant K at 298 K. [2]

Q37. 37. A* Challenge: Relating Entropy to Equilibrium
Constant $\Delta S_{\text{total}} = R \ln K$ 37

[5 Marks]
WCH14/01/Hard/Q37

For the reversible reaction 37: $\Delta S_{\text{system}} = -120 \text{ J K}^{-1} \text{ mol}^{-1}$, $\Delta H = -80.0 \text{ kJ mol}^{-1}$ at $T = 298 \text{ K}$.

- (a) Calculate $\Delta S_{\text{surroundings}}$ and ΔS_{total} at 298 K. [3]
(b) Use $\Delta S_{\text{total}} = R \ln K$ ($R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$) to calculate the numerical value of the equilibrium constant K at 298 K. [2]

Q38. 38. A* Challenge: Relating Entropy to Equilibrium
Constant $\Delta S_{\text{total}} = R \ln K$ 38

[5 Marks]
WCH14/01/Hard/Q38

For the reversible reaction 38: $\Delta S_{\text{system}} = -120 \text{ J K}^{-1} \text{ mol}^{-1}$, $\Delta H = -80.0 \text{ kJ mol}^{-1}$ at $T = 298 \text{ K}$.

- (a) Calculate $\Delta S_{\text{surroundings}}$ and ΔS_{total} at 298 K. [3]
(b) Use $\Delta S_{\text{total}} = R \ln K$ ($R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$) to calculate the numerical value of the equilibrium constant K at 298 K. [2]

Q39. 39. A* Challenge: Relating Entropy to Equilibrium
Constant $\Delta S_{\text{total}} = R \ln K$ 39

[5 Marks]
WCH14/01/Hard/Q39

For the reversible reaction 39: $\Delta S_{\text{system}} = -120 \text{ J K}^{-1} \text{ mol}^{-1}$, $\Delta H = -80.0 \text{ kJ mol}^{-1}$ at $T = 298 \text{ K}$.

- (a) Calculate $\Delta S_{\text{surroundings}}$ and ΔS_{total} at 298 K. [3]
(b) Use $\Delta S_{\text{total}} = R \ln K$ ($R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$) to calculate the numerical value of the equilibrium constant K at 298 K. [2]

Q40. 40. A* Challenge: Relating Entropy to Equilibrium
Constant $\Delta S_{\text{total}} = R \ln K$ 40

[5 Marks]
WCH14/01/Hard/Q40

For the reversible reaction 40: $\Delta S_{\text{system}} = -120 \text{ J K}^{-1} \text{ mol}^{-1}$, $\Delta H = -80.0 \text{ kJ mol}^{-1}$ at $T = 298 \text{ K}$.

- (a) Calculate $\Delta S_{\text{surroundings}}$ and ΔS_{total} at 298 K. [3]
(b) Use $\Delta S_{\text{total}} = R \ln K$ ($R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$) to calculate the numerical value of the equilibrium constant K at 298 K. [2]

Q41. 41. A* Challenge: Relating Entropy to Equilibrium
Constant $\Delta S_{\text{total}} = R \ln K$ 41

[5 Marks]
WCH14/01/Hard/Q41

For the reversible reaction 41: $\Delta S_{\text{system}} = -120 \text{ J K}^{-1} \text{ mol}^{-1}$, $\Delta H = -80.0 \text{ kJ mol}^{-1}$ at $T = 298 \text{ K}$.

- (a) Calculate $\Delta S_{\text{surroundings}}$ and ΔS_{total} at 298 K. [3]
(b) Use $\Delta S_{\text{total}} = R \ln K$ ($R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$) to calculate the numerical value of the equilibrium constant K at 298 K. [2]

Q42. 42. A* Challenge: Relating Entropy to Equilibrium
Constant $\Delta S_{\text{total}} = R \ln K$ 42

[5 Marks]
WCH14/01/Hard/Q42

For the reversible reaction 42: $\Delta S_{\text{system}} = -120 \text{ J K}^{-1} \text{ mol}^{-1}$, $\Delta H = -80.0 \text{ kJ mol}^{-1}$ at $T = 298 \text{ K}$.

- (a) Calculate $\Delta S_{\text{surroundings}}$ and ΔS_{total} at 298 K. [3]
(b) Use $\Delta S_{\text{total}} = R \ln K$ ($R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$) to calculate the numerical value of the equilibrium constant K at 298 K. [2]

Q43. 43. A* Challenge: Relating Entropy to Equilibrium
Constant $\Delta S_{\text{total}} = R \ln K$ 43

[5 Marks]
WCH14/01/Hard/Q43

For the reversible reaction 43: $\Delta S_{\text{system}} = -120 \text{ J K}^{-1} \text{ mol}^{-1}$, $\Delta H = -80.0 \text{ kJ mol}^{-1}$ at $T = 298 \text{ K}$.

- (a) Calculate $\Delta S_{\text{surroundings}}$ and ΔS_{total} at 298 K. [3]
(b) Use $\Delta S_{\text{total}} = R \ln K$ ($R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$) to calculate the numerical value of the equilibrium constant K at 298 K. [2]

Q44. 44. A* Challenge: Relating Entropy to Equilibrium
Constant $\Delta S_{\text{total}} = R \ln K$ 44

[5 Marks]
WCH14/01/Hard/Q44

For the reversible reaction 44: $\Delta S_{\text{system}} = -120 \text{ J K}^{-1} \text{ mol}^{-1}$, $\Delta H = -80.0 \text{ kJ mol}^{-1}$ at $T = 298 \text{ K}$.

- (a) Calculate $\Delta S_{\text{surroundings}}$ and ΔS_{total} at 298 K. [3]
(b) Use $\Delta S_{\text{total}} = R \ln K$ ($R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$) to calculate the numerical value of the equilibrium constant K at 298 K. [2]

Q45. 45. A* Challenge: Relating Entropy to Equilibrium
Constant $\Delta S_{\text{total}} = R \ln K$ 45

[5 Marks]
WCH14/01/Hard/Q45

For the reversible reaction 45: $\Delta S_{\text{system}} = -120 \text{ J K}^{-1} \text{ mol}^{-1}$, $\Delta H = -80.0 \text{ kJ mol}^{-1}$ at $T = 298 \text{ K}$.

- (a) Calculate $\Delta S_{\text{surroundings}}$ and ΔS_{total} at 298 K. [3]
(b) Use $\Delta S_{\text{total}} = R \ln K$ ($R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$) to calculate the numerical value of the equilibrium constant K at 298 K. [2]

Q46. 46. A* Challenge: Relating Entropy to Equilibrium
Constant $\Delta S_{\text{total}} = R \ln K$ 46

[5 Marks]
WCH14/01/Hard/Q46

For the reversible reaction 46: $\Delta S_{\text{system}} = -120 \text{ J K}^{-1} \text{ mol}^{-1}$, $\Delta H = -80.0 \text{ kJ mol}^{-1}$ at $T = 298 \text{ K}$.

- (a) Calculate $\Delta S_{\text{surroundings}}$ and ΔS_{total} at 298 K. [3]
(b) Use $\Delta S_{\text{total}} = R \ln K$ ($R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$) to calculate the numerical value of the equilibrium constant K at 298 K. [2]

Q47. 47. A* Challenge: Relating Entropy to Equilibrium**[5 Marks]****Constant $\Delta S_{\text{total}} = R \ln K$ 47****WCH14/01/Hard/Q47**

For the reversible reaction 47: $\Delta S_{\text{system}} = -120 \text{ J K}^{-1} \text{ mol}^{-1}$, $\Delta H = -80.0 \text{ kJ mol}^{-1}$ at $T = 298 \text{ K}$.

- (a) Calculate $\Delta S_{\text{surroundings}}$ and ΔS_{total} at 298 K. [3]
(b) Use $\Delta S_{\text{total}} = R \ln K$ ($R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$) to calculate the numerical value of the equilibrium constant K at 298 K. [2]

Q48. 48. A* Challenge: Relating Entropy to Equilibrium**[5 Marks]****Constant $\Delta S_{\text{total}} = R \ln K$ 48****WCH14/01/Hard/Q48**

For the reversible reaction 48: $\Delta S_{\text{system}} = -120 \text{ J K}^{-1} \text{ mol}^{-1}$, $\Delta H = -80.0 \text{ kJ mol}^{-1}$ at $T = 298 \text{ K}$.

- (a) Calculate $\Delta S_{\text{surroundings}}$ and ΔS_{total} at 298 K. [3]
(b) Use $\Delta S_{\text{total}} = R \ln K$ ($R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$) to calculate the numerical value of the equilibrium constant K at 298 K. [2]

Q49. 49. A* Challenge: Relating Entropy to Equilibrium**[5 Marks]****Constant $\Delta S_{\text{total}} = R \ln K$ 49****WCH14/01/Hard/Q49**

For the reversible reaction 49: $\Delta S_{\text{system}} = -120 \text{ J K}^{-1} \text{ mol}^{-1}$, $\Delta H = -80.0 \text{ kJ mol}^{-1}$ at $T = 298 \text{ K}$.

- (a) Calculate $\Delta S_{\text{surroundings}}$ and ΔS_{total} at 298 K. [3]
(b) Use $\Delta S_{\text{total}} = R \ln K$ ($R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$) to calculate the numerical value of the equilibrium constant K at 298 K. [2]

Q50. 50. A* Challenge: Relating Entropy to Equilibrium**[5 Marks]****Constant $\Delta S_{\text{total}} = R \ln K$ 50****WCH14/01/Hard/Q50**

For the reversible reaction 50: $\Delta S_{\text{system}} = -120 \text{ J K}^{-1} \text{ mol}^{-1}$, $\Delta H = -80.0 \text{ kJ mol}^{-1}$ at $T = 298 \text{ K}$.

- (a) Calculate $\Delta S_{\text{surroundings}}$ and ΔS_{total} at 298 K. [3]
(b) Use $\Delta S_{\text{total}} = R \ln K$ ($R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$) to calculate the numerical value of the equilibrium constant K at 298 K. [2]

SECTION C: WORKED MARK SCHEME (Q1–Q50) — 13A

Q	MARKING SCHEME & ACCEPTABLE ANSWERS	MARKS
Q1	Q1: 1. Equilibrium Constant Kc Expression and Units 1. (a) $K_c = \frac{[\text{CH}_3\text{COOC}_2\text{H}_5][\text{H}_2\text{O}]}{([\text{CH}_3\text{COOH}][\text{C}_2\text{H}_5\text{OH}])}$ (1). 1. (b) No units (units cancel out) (1). 1. (c) Eqm moles: acid=0.33, alcohol=0.33, ester=0.67, water=0.67. $K_c = \frac{(0.67 \times 0.67)}{(0.33 \times 0.33)} = 4.12$ (2).	[4]
Q2	Q2: 2. Gas Phase Equilibrium Kp Calculation 2. (a) $p(\text{N}_2) = 0.20 \times 100 = 20 \text{ atm}$, $p(\text{H}_2) = 0.60 \times 100 = 60 \text{ atm}$, $p(\text{NH}_3) = 0.20 \times 100 = 20 \text{ atm}$ (2). 2. (b) $K_p = \frac{p(\text{NH}_3)^2}{(p(\text{N}_2) \times p(\text{H}_2)^3)}$ (1). $K_p = \frac{(20)^2}{((20) \times (60)^3)} = 400 / 4320000 = 9.26 \times 10^{-5} \text{ atm}^{-2}$ (2).	[5]
Q3	Q3: 3. Equilibrium Calculation & Factors Step 3 3. (a) Kc decreases (1). Forward reaction is exothermic; increasing temp shifts equilibrium left to absorb heat (1). 3. (b) No effect on Kc (only temperature changes Kc) (1).	[3]
Q4	Q4: 4. Equilibrium Calculation & Factors Step 4 4. (a) Kc decreases (1). Forward reaction is exothermic; increasing temp shifts equilibrium left to absorb heat (1). 4. (b) No effect on Kc (only temperature changes Kc) (1).	[3]
Q5	Q5: 5. Equilibrium Calculation & Factors Step 5 5. (a) Kc decreases (1). Forward reaction is exothermic; increasing temp shifts equilibrium left to absorb heat (1). 5. (b) No effect on Kc (only temperature changes Kc) (1).	[3]
Q6	Q6: 6. Equilibrium Calculation & Factors Step 6 6. (a) Kc decreases (1). Forward reaction is exothermic; increasing temp shifts equilibrium left to absorb heat (1). 6. (b) No effect on Kc (only temperature changes Kc) (1).	[3]
Q7	Q7: 7. Equilibrium Calculation & Factors Step 7 7. (a) Kc decreases (1). Forward reaction is exothermic; increasing temp shifts equilibrium left to absorb heat (1). 7. (b) No effect on Kc (only temperature changes Kc) (1).	[3]
Q8	Q8: 8. Equilibrium Calculation & Factors Step 8 8. (a) Kc decreases (1). Forward reaction is exothermic; increasing temp shifts equilibrium left to absorb heat (1). 8. (b) No effect on Kc (only temperature changes Kc) (1).	[3]
Q9	Q9: 9. Equilibrium Calculation & Factors Step 9 9. (a) Kc decreases (1). Forward reaction is exothermic; increasing temp shifts equilibrium left to absorb heat (1). 9. (b) No effect on Kc (only temperature changes Kc) (1).	[3]
Q10	Q10: 10. Equilibrium Calculation & Factors Step 10 10. (a) Kc decreases (1). Forward reaction is exothermic; increasing temp shifts equilibrium left to absorb heat (1). 10. (b) No effect on Kc (only temperature changes Kc) (1).	[3]
Q11	Q11: 11. Equilibrium Calculation & Factors Step 11 11. (a) Kc decreases (1). Forward reaction is exothermic; increasing temp shifts equilibrium left to absorb heat (1). 11. (b) No effect on Kc (only temperature changes Kc) (1).	[3]
Q12	Q12: 12. Equilibrium Calculation & Factors Step 12 12. (a) Kc decreases (1). Forward reaction is exothermic; increasing temp shifts equilibrium left to absorb heat (1). 12. (b) No effect on Kc (only temperature changes Kc) (1).	[3]

Q13	Q13: 13. Equilibrium Calculation & Factors Step 13 13. (a) K_c decreases (1). Forward reaction is exothermic; increasing temp shifts equilibrium left to absorb heat (1). 13. (b) No effect on K_c (only temperature changes K_c) (1).	[3]
Q14	Q14: 14. Equilibrium Calculation & Factors Step 14 14. (a) K_c decreases (1). Forward reaction is exothermic; increasing temp shifts equilibrium left to absorb heat (1). 14. (b) No effect on K_c (only temperature changes K_c) (1).	[3]
Q15	Q15: 15. Equilibrium Calculation & Factors Step 15 15. (a) K_c decreases (1). Forward reaction is exothermic; increasing temp shifts equilibrium left to absorb heat (1). 15. (b) No effect on K_c (only temperature changes K_c) (1).	[3]
Q16	Q16: 16. Equilibrium Calculation & Factors Step 16 16. (a) K_c decreases (1). Forward reaction is exothermic; increasing temp shifts equilibrium left to absorb heat (1). 16. (b) No effect on K_c (only temperature changes K_c) (1).	[3]
Q17	Q17: 17. Equilibrium Calculation & Factors Step 17 17. (a) K_c decreases (1). Forward reaction is exothermic; increasing temp shifts equilibrium left to absorb heat (1). 17. (b) No effect on K_c (only temperature changes K_c) (1).	[3]
Q18	Q18: 18. Equilibrium Calculation & Factors Step 18 18. (a) K_c decreases (1). Forward reaction is exothermic; increasing temp shifts equilibrium left to absorb heat (1). 18. (b) No effect on K_c (only temperature changes K_c) (1).	[3]
Q19	Q19: 19. Equilibrium Calculation & Factors Step 19 19. (a) K_c decreases (1). Forward reaction is exothermic; increasing temp shifts equilibrium left to absorb heat (1). 19. (b) No effect on K_c (only temperature changes K_c) (1).	[3]
Q20	Q20: 20. Equilibrium Calculation & Factors Step 20 20. (a) K_c decreases (1). Forward reaction is exothermic; increasing temp shifts equilibrium left to absorb heat (1). 20. (b) No effect on K_c (only temperature changes K_c) (1).	[3]
Q21	Q21: 21. Equilibrium Calculation & Factors Step 21 21. (a) K_c decreases (1). Forward reaction is exothermic; increasing temp shifts equilibrium left to absorb heat (1). 21. (b) No effect on K_c (only temperature changes K_c) (1).	[3]
Q22	Q22: 22. Equilibrium Calculation & Factors Step 22 22. (a) K_c decreases (1). Forward reaction is exothermic; increasing temp shifts equilibrium left to absorb heat (1). 22. (b) No effect on K_c (only temperature changes K_c) (1).	[3]
Q23	Q23: 23. Equilibrium Calculation & Factors Step 23 23. (a) K_c decreases (1). Forward reaction is exothermic; increasing temp shifts equilibrium left to absorb heat (1). 23. (b) No effect on K_c (only temperature changes K_c) (1).	[3]
Q24	Q24: 24. Equilibrium Calculation & Factors Step 24 24. (a) K_c decreases (1). Forward reaction is exothermic; increasing temp shifts equilibrium left to absorb heat (1). 24. (b) No effect on K_c (only temperature changes K_c) (1).	[3]
Q25	Q25: 25. Equilibrium Calculation & Factors Step 25 25. (a) K_c decreases (1). Forward reaction is exothermic; increasing temp shifts equilibrium left to absorb heat (1). 25. (b) No effect on K_c (only temperature changes K_c) (1).	[3]

Q26	Q26: 26. A* Challenge: Relating Entropy to Equilibrium Constant $\Delta S_{\text{total}} = R \ln K$ 26 26. (a) $\Delta S_{\text{surr}} = -(-80000) / 298 = +268.5 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). $\Delta S_{\text{tot}} = -120 + 268.5 = +148.5 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). 26. (b) $\ln K = \Delta S_{\text{tot}} / R = 148.5 / 8.31 = 17.87$ (1). $K = e^{17.87} = 5.76 \times 10^7$ (1).	[5]
Q27	Q27: 27. A* Challenge: Relating Entropy to Equilibrium Constant $\Delta S_{\text{total}} = R \ln K$ 27 27. (a) $\Delta S_{\text{surr}} = -(-80000) / 298 = +268.5 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). $\Delta S_{\text{tot}} = -120 + 268.5 = +148.5 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). 27. (b) $\ln K = \Delta S_{\text{tot}} / R = 148.5 / 8.31 = 17.87$ (1). $K = e^{17.87} = 5.76 \times 10^7$ (1).	[5]
Q28	Q28: 28. A* Challenge: Relating Entropy to Equilibrium Constant $\Delta S_{\text{total}} = R \ln K$ 28 28. (a) $\Delta S_{\text{surr}} = -(-80000) / 298 = +268.5 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). $\Delta S_{\text{tot}} = -120 + 268.5 = +148.5 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). 28. (b) $\ln K = \Delta S_{\text{tot}} / R = 148.5 / 8.31 = 17.87$ (1). $K = e^{17.87} = 5.76 \times 10^7$ (1).	[5]
Q29	Q29: 29. A* Challenge: Relating Entropy to Equilibrium Constant $\Delta S_{\text{total}} = R \ln K$ 29 29. (a) $\Delta S_{\text{surr}} = -(-80000) / 298 = +268.5 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). $\Delta S_{\text{tot}} = -120 + 268.5 = +148.5 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). 29. (b) $\ln K = \Delta S_{\text{tot}} / R = 148.5 / 8.31 = 17.87$ (1). $K = e^{17.87} = 5.76 \times 10^7$ (1).	[5]
Q30	Q30: 30. A* Challenge: Relating Entropy to Equilibrium Constant $\Delta S_{\text{total}} = R \ln K$ 30 30. (a) $\Delta S_{\text{surr}} = -(-80000) / 298 = +268.5 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). $\Delta S_{\text{tot}} = -120 + 268.5 = +148.5 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). 30. (b) $\ln K = \Delta S_{\text{tot}} / R = 148.5 / 8.31 = 17.87$ (1). $K = e^{17.87} = 5.76 \times 10^7$ (1).	[5]
Q31	Q31: 31. A* Challenge: Relating Entropy to Equilibrium Constant $\Delta S_{\text{total}} = R \ln K$ 31 31. (a) $\Delta S_{\text{surr}} = -(-80000) / 298 = +268.5 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). $\Delta S_{\text{tot}} = -120 + 268.5 = +148.5 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). 31. (b) $\ln K = \Delta S_{\text{tot}} / R = 148.5 / 8.31 = 17.87$ (1). $K = e^{17.87} = 5.76 \times 10^7$ (1).	[5]
Q32	Q32: 32. A* Challenge: Relating Entropy to Equilibrium Constant $\Delta S_{\text{total}} = R \ln K$ 32 32. (a) $\Delta S_{\text{surr}} = -(-80000) / 298 = +268.5 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). $\Delta S_{\text{tot}} = -120 + 268.5 = +148.5 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). 32. (b) $\ln K = \Delta S_{\text{tot}} / R = 148.5 / 8.31 = 17.87$ (1). $K = e^{17.87} = 5.76 \times 10^7$ (1).	[5]
Q33	Q33: 33. A* Challenge: Relating Entropy to Equilibrium Constant $\Delta S_{\text{total}} = R \ln K$ 33 33. (a) $\Delta S_{\text{surr}} = -(-80000) / 298 = +268.5 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). $\Delta S_{\text{tot}} = -120 + 268.5 = +148.5 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). 33. (b) $\ln K = \Delta S_{\text{tot}} / R = 148.5 / 8.31 = 17.87$ (1). $K = e^{17.87} = 5.76 \times 10^7$ (1).	[5]
Q34	Q34: 34. A* Challenge: Relating Entropy to Equilibrium Constant $\Delta S_{\text{total}} = R \ln K$ 34 34. (a) $\Delta S_{\text{surr}} = -(-80000) / 298 = +268.5 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). $\Delta S_{\text{tot}} = -120 + 268.5 = +148.5 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). 34. (b) $\ln K = \Delta S_{\text{tot}} / R = 148.5 / 8.31 = 17.87$ (1). $K = e^{17.87} = 5.76 \times 10^7$ (1).	[5]
Q35	Q35: 35. A* Challenge: Relating Entropy to Equilibrium Constant $\Delta S_{\text{total}} = R \ln K$ 35 35. (a) $\Delta S_{\text{surr}} = -(-80000) / 298 = +268.5 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). $\Delta S_{\text{tot}} = -120 + 268.5 = +148.5 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). 35. (b) $\ln K = \Delta S_{\text{tot}} / R = 148.5 / 8.31 = 17.87$ (1). $K = e^{17.87} = 5.76 \times 10^7$ (1).	[5]
Q36	Q36: 36. A* Challenge: Relating Entropy to Equilibrium Constant $\Delta S_{\text{total}} = R \ln K$ 36 36. (a) $\Delta S_{\text{surr}} = -(-80000) / 298 = +268.5 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). $\Delta S_{\text{tot}} = -120 + 268.5 = +148.5 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). 36. (b) $\ln K = \Delta S_{\text{tot}} / R = 148.5 / 8.31 = 17.87$ (1). $K = e^{17.87} = 5.76 \times 10^7$ (1).	[5]
Q37	Q37: 37. A* Challenge: Relating Entropy to Equilibrium Constant $\Delta S_{\text{total}} = R \ln K$ 37 37. (a) $\Delta S_{\text{surr}} = -(-80000) / 298 = +268.5 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). $\Delta S_{\text{tot}} = -120 + 268.5 = +148.5 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). 37. (b) $\ln K = \Delta S_{\text{tot}} / R = 148.5 / 8.31 = 17.87$ (1). $K = e^{17.87} = 5.76 \times 10^7$ (1).	[5]
Q38	Q38: 38. A* Challenge: Relating Entropy to Equilibrium Constant $\Delta S_{\text{total}} = R \ln K$ 38 38. (a) $\Delta S_{\text{surr}} = -(-80000) / 298 = +268.5 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). $\Delta S_{\text{tot}} = -120 + 268.5 = +148.5 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). 38. (b) $\ln K = \Delta S_{\text{tot}} / R = 148.5 / 8.31 = 17.87$ (1). $K = e^{17.87} = 5.76 \times 10^7$ (1).	[5]

Q39	Q39: 39. A* Challenge: Relating Entropy to Equilibrium Constant $\Delta S_{\text{total}} = R \ln K$ 39 39. (a) $\Delta S_{\text{surr}} = -(-80000) / 298 = +268.5 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). $\Delta S_{\text{tot}} = -120 + 268.5 = +148.5 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). 39. (b) $\ln K = \Delta S_{\text{tot}} / R = 148.5 / 8.31 = 17.87$ (1). $K = e^{17.87} = 5.76 \times 10^7$ (1).	[5]
Q40	Q40: 40. A* Challenge: Relating Entropy to Equilibrium Constant $\Delta S_{\text{total}} = R \ln K$ 40 40. (a) $\Delta S_{\text{surr}} = -(-80000) / 298 = +268.5 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). $\Delta S_{\text{tot}} = -120 + 268.5 = +148.5 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). 40. (b) $\ln K = \Delta S_{\text{tot}} / R = 148.5 / 8.31 = 17.87$ (1). $K = e^{17.87} = 5.76 \times 10^7$ (1).	[5]
Q41	Q41: 41. A* Challenge: Relating Entropy to Equilibrium Constant $\Delta S_{\text{total}} = R \ln K$ 41 41. (a) $\Delta S_{\text{surr}} = -(-80000) / 298 = +268.5 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). $\Delta S_{\text{tot}} = -120 + 268.5 = +148.5 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). 41. (b) $\ln K = \Delta S_{\text{tot}} / R = 148.5 / 8.31 = 17.87$ (1). $K = e^{17.87} = 5.76 \times 10^7$ (1).	[5]
Q42	Q42: 42. A* Challenge: Relating Entropy to Equilibrium Constant $\Delta S_{\text{total}} = R \ln K$ 42 42. (a) $\Delta S_{\text{surr}} = -(-80000) / 298 = +268.5 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). $\Delta S_{\text{tot}} = -120 + 268.5 = +148.5 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). 42. (b) $\ln K = \Delta S_{\text{tot}} / R = 148.5 / 8.31 = 17.87$ (1). $K = e^{17.87} = 5.76 \times 10^7$ (1).	[5]
Q43	Q43: 43. A* Challenge: Relating Entropy to Equilibrium Constant $\Delta S_{\text{total}} = R \ln K$ 43 43. (a) $\Delta S_{\text{surr}} = -(-80000) / 298 = +268.5 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). $\Delta S_{\text{tot}} = -120 + 268.5 = +148.5 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). 43. (b) $\ln K = \Delta S_{\text{tot}} / R = 148.5 / 8.31 = 17.87$ (1). $K = e^{17.87} = 5.76 \times 10^7$ (1).	[5]
Q44	Q44: 44. A* Challenge: Relating Entropy to Equilibrium Constant $\Delta S_{\text{total}} = R \ln K$ 44 44. (a) $\Delta S_{\text{surr}} = -(-80000) / 298 = +268.5 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). $\Delta S_{\text{tot}} = -120 + 268.5 = +148.5 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). 44. (b) $\ln K = \Delta S_{\text{tot}} / R = 148.5 / 8.31 = 17.87$ (1). $K = e^{17.87} = 5.76 \times 10^7$ (1).	[5]
Q45	Q45: 45. A* Challenge: Relating Entropy to Equilibrium Constant $\Delta S_{\text{total}} = R \ln K$ 45 45. (a) $\Delta S_{\text{surr}} = -(-80000) / 298 = +268.5 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). $\Delta S_{\text{tot}} = -120 + 268.5 = +148.5 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). 45. (b) $\ln K = \Delta S_{\text{tot}} / R = 148.5 / 8.31 = 17.87$ (1). $K = e^{17.87} = 5.76 \times 10^7$ (1).	[5]
Q46	Q46: 46. A* Challenge: Relating Entropy to Equilibrium Constant $\Delta S_{\text{total}} = R \ln K$ 46 46. (a) $\Delta S_{\text{surr}} = -(-80000) / 298 = +268.5 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). $\Delta S_{\text{tot}} = -120 + 268.5 = +148.5 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). 46. (b) $\ln K = \Delta S_{\text{tot}} / R = 148.5 / 8.31 = 17.87$ (1). $K = e^{17.87} = 5.76 \times 10^7$ (1).	[5]
Q47	Q47: 47. A* Challenge: Relating Entropy to Equilibrium Constant $\Delta S_{\text{total}} = R \ln K$ 47 47. (a) $\Delta S_{\text{surr}} = -(-80000) / 298 = +268.5 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). $\Delta S_{\text{tot}} = -120 + 268.5 = +148.5 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). 47. (b) $\ln K = \Delta S_{\text{tot}} / R = 148.5 / 8.31 = 17.87$ (1). $K = e^{17.87} = 5.76 \times 10^7$ (1).	[5]
Q48	Q48: 48. A* Challenge: Relating Entropy to Equilibrium Constant $\Delta S_{\text{total}} = R \ln K$ 48 48. (a) $\Delta S_{\text{surr}} = -(-80000) / 298 = +268.5 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). $\Delta S_{\text{tot}} = -120 + 268.5 = +148.5 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). 48. (b) $\ln K = \Delta S_{\text{tot}} / R = 148.5 / 8.31 = 17.87$ (1). $K = e^{17.87} = 5.76 \times 10^7$ (1).	[5]
Q49	Q49: 49. A* Challenge: Relating Entropy to Equilibrium Constant $\Delta S_{\text{total}} = R \ln K$ 49 49. (a) $\Delta S_{\text{surr}} = -(-80000) / 298 = +268.5 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). $\Delta S_{\text{tot}} = -120 + 268.5 = +148.5 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). 49. (b) $\ln K = \Delta S_{\text{tot}} / R = 148.5 / 8.31 = 17.87$ (1). $K = e^{17.87} = 5.76 \times 10^7$ (1).	[5]
Q50	Q50: 50. A* Challenge: Relating Entropy to Equilibrium Constant $\Delta S_{\text{total}} = R \ln K$ 50 50. (a) $\Delta S_{\text{surr}} = -(-80000) / 298 = +268.5 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). $\Delta S_{\text{tot}} = -120 + 268.5 = +148.5 \text{ J K}^{-1} \text{ mol}^{-1}$ (1). 50. (b) $\ln K = \Delta S_{\text{tot}} / R = 148.5 / 8.31 = 17.87$ (1). $K = e^{17.87} = 5.76 \times 10^7$ (1).	[5]

SECTION D: EMPIRICAL EXAMINER FAQs & COMMON TRAPS (10 FAQs) — 13A

FAQ #1: Square Brackets vs Round Brackets in Kc

Category: Notation Trap

COMMON EXAMINER TRAP / MISTAKE:

Using round brackets () instead of square brackets [] in Kc expressions.

MODEL ANSWER / EXAMINER EXPECTATION:

Kc MUST use square brackets [X] denoting concentration in mol dm⁻³. Using round brackets (X) loses the mark.

FAQ #2: Solids and Liquids in Heterogeneous Equilibria

Category: Heterogeneous K

COMMON EXAMINER TRAP / MISTAKE:

Including solid or pure liquid concentrations in Kc expressions.

MODEL ANSWER / EXAMINER EXPECTATION:

Heterogeneous equilibria: Solids and pure liquid solvent concentrations remain constant and are omitted from Kc (e.g. $\text{CaCO}_3(\text{s}) \rightarrow \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$ has $K_c = [\text{CO}_2]$).

FAQ #3: Mole Fraction vs Partial Pressure

Category: Kp Definitions

COMMON EXAMINER TRAP / MISTAKE:

Confusing mole fraction x_A with partial pressure p_A .

MODEL ANSWER / EXAMINER EXPECTATION:

Mole fraction $x_A = \text{moles of A} / \text{total moles}$. Partial pressure $p_A = x_A \times P_{\text{total}}$. Kp MUST use partial pressures p_A .

FAQ #4: Why Pressure Does Not Change Kp

Category: Equilibrium Factors

COMMON EXAMINER TRAP / MISTAKE:

Stating that increasing pressure increases Kp for gas reactions.

MODEL ANSWER / EXAMINER EXPECTATION:

Increasing pressure increases individual partial pressures, but the ratio in Kp remains CONSTANT because the equilibrium composition shifts to restore Kp. ONLY TEMPERATURE changes Kp/Kc.

FAQ #5: Why Catalysts Do Not Shift Equilibrium

Category: Catalysts & K

COMMON EXAMINER TRAP / MISTAKE:

Claiming a catalyst increases equilibrium yield or changes K.

MODEL ANSWER / EXAMINER EXPECTATION:

A catalyst increases the rate of BOTH forward and reverse reactions equally by lowering E_a for both routes. It speeds up attainment of equilibrium but does NOT alter equilibrium position or K.

FAQ #6: Temperature Effect on Exothermic Reactions

Category: Temperature Dependence

COMMON EXAMINER TRAP / MISTAKE:

Stating Kc increases with temperature for exothermic reactions.

MODEL ANSWER / EXAMINER EXPECTATION:

Exothermic reaction: ΔH negative. Increasing temp causes $\Delta S_{\text{surroundings}}$ ($-\Delta H / T$) to become less positive $\rightarrow \Delta S_{\text{total}}$ decreases $\rightarrow K$ DECREASES.

FAQ #7: Temperature Effect on Endothermic Reactions**Category: Temperature Dependence****COMMON EXAMINER TRAP / MISTAKE:**

Stating K_c decreases with temperature for endothermic reactions.

MODEL ANSWER / EXAMINER EXPECTATION:

Endothermic reaction: ΔH positive. Increasing temp causes $\Delta S_{\text{surroundings}}$ ($-\Delta H / T$) to become less negative $\rightarrow \Delta S_{\text{total}}$ increases $\rightarrow K$ INCREASES.

FAQ #8: Units of K_p **Category: K_p Units****COMMON EXAMINER TRAP / MISTAKE:**

Assuming K_p is always unitless.

MODEL ANSWER / EXAMINER EXPECTATION:

Units of K_p depend on the stoichiometry: substitute atm or Pa into K_p expression. E.g. $K_p = \frac{p(\text{NH}_3)^2}{p(\text{N}_2)p(\text{H}_2)^3}$ has units atm^{-2} or Pa^{-2} .

FAQ #9: Deriving K from ΔS_{total} **Category: Thermodynamics to K** **COMMON EXAMINER TRAP / MISTAKE:**

Using $\log_{10}$ instead of natural log ($\ln$) in $\Delta S_{\text{total}} = R \ln K$.

MODEL ANSWER / EXAMINER EXPECTATION:

$\Delta S_{\text{total}} = R \ln K$ uses NATURAL LOG (base e). To solve for K : $K = e^{(\Delta S_{\text{total}} / R)}$. Using $10^{(\Delta S / R)}$ is wrong.

FAQ #10: Large vs Small K Value Significance**Category: Equilibrium Position****COMMON EXAMINER TRAP / MISTAKE:**

Confusing $K \gg 1$ with fast reaction rate.

MODEL ANSWER / EXAMINER EXPECTATION:

$K \gg 1$ (e.g. $K > 10^3$) means equilibrium lies far to the RIGHT (high product yield at equilibrium). It says NOTHING about reaction rate (which depends on activation energy).